PWM Generation for LED Brightness Control Using STM32

This project demonstrates how to generate a PWM signal using the STM32 Timer peripheral to control the brightness of an LED.

The PWM signal is generated using the STM32 HAL library and configured through STM32CubeMX/ STM32CubeIDE.

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1. STM32CubeMX / IOC Configuration

Select Timer

Open the `.ioc` file in STM32CubeMX or STM32CubeIDE.

Select the required Timer peripheral.

Example:

```text id="h5j8r0" TIM2
```

```
### Configure Clock Source

Select the appropriate clock source depending on the STM32 board and clock configuration.

```text id="0uh1c3"
Clock Source → Internal Clock / External Clock
```

Configure PWM Channel

Select the required Timer channel and configure it for PWM generation.

Example:

```text id="5om7xy" Channel → PWM Generation CH1
```

```
For example:
```
```

```text id="q3w2mj"
TIM2 → Channel 1 → PWM Generation CH1
```

The actual Timer and PWM output pin depend on the STM32 microcontroller and hardware configuration.

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2. Timer Parameter Settings

The PWM frequency determines how fast the LED is switched ON and OFF.

The PWM period is calculated using:

```text id="a8f8y4" T = 1 / f
```

```
For example, if the desired PWM frequency is **1 kHz**:

```text id="u8m2gu"
T = 1 / 1000
T = 0.001 seconds
T = 1 ms
```

If the timer is configured with a **1 µs timer tick**, the Counter Period can be:

```text id="1g5l5f" Counter Period = 1000 - 1 = 999
```

```
Therefore:

```text id="u2zv4d"
Prescaler       → Timer Clock / 1 MHz - 1
Counter Mode    → Up
Counter Period  → 999
Auto Reload     → Enable
PWM Mode        → PWM Mode 1
Pulse           → Initial Compare Value
```

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3. PWM Duty Cycle and LED Brightness

The PWM duty cycle controls the average power supplied to the LED and therefore controls its apparent brightness.

With:

```text id="7o2n6j" ARR = 999
```

```
the Compare value can be changed to control brightness.

### 100% Brightness

```c id="u6k4p0"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 999);
```

Approximate duty cycle:

```text id="5j5q7f" 100%
```

```
### 75% Brightness

```c id="x8j3h2"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 750);
```

Approximate duty cycle:

```text id="p8q6j2" 75%
```

```
### 50% Brightness

```c id="k1j5m0"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 500);
```

Approximate duty cycle:

```text id="3l9h7z" 50%
```

```
### 25% Brightness

```c id="b6v2c8"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 250);
```

Approximate duty cycle:

```text id="j9x4n3" 25%

```
0% Brightness

```c id="q7m2k5"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 0);
```

Approximate duty cycle:

```text id="t6n4p2" 0%

```
The relationship is approximately:

```text id="5qv3x1"
Compare Value
      │
      ▼
PWM Duty Cycle
      │
      ▼
Average LED Power
      │
      ▼
LED Brightness
```

---

4. PWM Generation in `main.c`

Start PWM generation using:

```c id="j7v2k4" HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);

```
This starts PWM output on:

```text id="0j7p9s"
TIM2 Channel 1
```

The Timer and Channel depend on the configuration selected in the `.ioc` file.

Change LED Brightness

The PWM duty cycle can be changed using:

```c id="3m5v8k" __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, VALUE);

```
For example:

```c id="h4x8n2"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 999);
```

sets approximately:

```text id="x8c3m1" 100% brightness

```
Similarly:

```c id="f5q2z7"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 500);
```

sets approximately:

```text id="j8m4k6" 50% brightness

```
---

5. Complete PWM LED Example

The following example continuously changes the LED brightness:

```c id="a6n3w8"
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);

while (1)
{
    /* 100% brightness */

    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 999);

    HAL_Delay(2000);


    /* 75% brightness */
```

```
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 750);
    HAL_Delay(2000);

    /* 50% brightness */
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 500);
    HAL_Delay(2000);

    /* 25% brightness */
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 250);
    HAL_Delay(2000);

    /* 0% brightness */
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 0);
    HAL_Delay(2000);
}
```

The LED brightness changes in the following sequence:

```text id="r3c6n9" 100% ↓ 75% ↓ 50% ↓ 25% ↓ 0% ↓ Repeat

```
---

6. Check PWM Configuration

If the LED does not glow or the PWM output does not work, check the following
file:

```text id="0q5z7m"
stm32f4xx_hal_msp.c
```

Verify that the required Timer peripheral and GPIO pin are initialized correctly.

Check:

- Timer clock is enabled.
- GPIO clock is enabled.
- PWM GPIO pin is initialized.
- GPIO is configured as **Alternate Function**.
- Correct Alternate Function is selected.
- Correct Timer Channel is selected.
- Physical LED pin matches the configured Timer Channel.
- `HAL_TIM_PWM_Start()` is called.
- Timer Prescaler is correct.
- Counter Period is correct.
- PWM Mode is set to PWM Mode 1.

The PWM GPIO should typically be configured as:

```text id="q8m3r1"
GPIO Mode → Alternate Function
Alternate Function → Correct TIMx_CHy
```

```
---

# 7. PWM LED Summary

Example Timer configuration:

```text id="c5v9x2"
PWM Frequency  = 1 kHz
Timer Tick     = 1 µs
Counter Period = 999
```
```

Duty cycle control:

```text id="8h3k5m"
Compare Value = 0 → 0% → LED OFF
Compare Value = 250 → 25% → Low brightness
Compare Value = 500 → 50% → Medium brightness
Compare Value = 750 → 75% → High brightness
Compare Value = 999 → ~100% → Maximum brightness
```

```
The PWM output is controlled using:

```c id="p4w8m2"
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, VALUE);
```
```

Overall operation:

```text id="q7x2m9"
STM32
```

```
  |
  | Timer Configuration
  ▼
PWM Generation
  |
  | Duty Cycle
  ▼
PWM Output Pin
  |
  ▼
LED
  |
  ▼
Brightness Control
```

> **Note:** The perceived brightness of an LED is not always perfectly linear with PWM duty cycle because human vision has a nonlinear response. For advanced LED dimming, gamma correction can be used.